Materials Today: Proceedings 70 (2022) 687–693

Contents lists available at ScienceDirect

Materials Today: Proceedings

journal homepage: www.elsevier.com/locate/matpr

# Integrating parametric design and additive manufacturing knowledge in industrial design education

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A R T I C L E   I N F O

*Article history:*
Available online 22 October 2022

*Keywords:*
Design for Additive Manufacturing (DfAM)
Design rules
Design guidelines
Industrial design engineering curriculum

A B S T R A C T

Additive Manufacturing (AM) technologies are present in almost any industry. For industrial designers, AM technologies are important for prototyping of new design concepts, and for their evaluation in short time and with less effort. Physical prototypes in industrial design often represent the outer shape, main modules and basic motions proposed with the designed concept. They are useful for better communication and acceptance of the design. Efficient prototyping can support creativity and experimentation in design.

However, AM technologies also have limitations, meaning that there are manufacturing restrictions that the designer needs to know. Lack of knowledge on DfAM tools and techniques can be seen as one of the barriers for the further implementation of AM. By implementing DfAM in the Industrial Design Engineering curriculum, future designers can work confidently with AM and design products that fully exploit the possibilities of these technologies. DfAM is a way of thinking, a way of designing that the future designers and engineers need to obtain in the educational process.

In this paper, we present the learning process that connects parametric modelling, DfAM and AM technologies through practical examples of product modelling and prototyping. The product models are redesigned and physically prototyped by the students, using a selected AM technology. More advanced examples are introduced through student projects that can stimulate creativity in development of design ideas, based on improved understanding of how the objects are manufactured, to get more advantage from using AM technologies.


## 1. Introduction

The development of additive manufacturing (AM) technologies, in correlation with parametric modeling, have opened new possibilities for creativity, aesthetics and efficiency in the industrial design process. More demanding designs can be developed, prototyped and fabricated. To benefit from the concurrent application of these technologies in the design process, industrial designers should be educated and trained for parametric design thinking and cooperation with computer programmers and production technology experts. On the other hand, the design process itself is highly dependent on personal creativity, sensory perception of form and material, and understanding of consumer trends.

The perceived complexity has an influence on determining the structure of the teaching methodology used to educate young designers of how to use these technologic achievements in product design and industrial design process. The motivation for learning comes from the fact that the customers often intuitively relate the aesthetic characteristics of the design to the functional quality of the product, and also, the fascination with the mathematical perfection and complexity of the form and its details. Parametric design has become a trend in many areas, such as architecture [1], fashion, graphic design and even food design.

The emphasis of the learning process presented in this paper is on extending the knowledge and practical experience for connecting the parametric design and additive production technologies through design for additive manufacturing (DfAM). Generally, AM processes enhance the design freedom in creating form, resulting in new opportunities for improved aesthetics and functionality, compared to the traditional manufacturing processes. For example,

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https://doi.org/10.1016/j.matpr.2022.10.124
2214-7853/

there are no limitations on shape imposed by the tool approach to the machined surface. When properly designed for a particular AM technology, complex parts can be prototyped and/or fabricated directly. In addition, production costs of AM are not related to the geometric complexity of the part [2], as for conventional material removal processes.

## 2. AM technologies and parametric design

AM technologies have application in diverse areas, including prototyping, manufacturing of specific parts, personalized medical devices, and molds. AM machines are affordable and popular for creating and evaluating design prototypes for appearance, ergonomics, proportions, functionality, and design communication. These technologies have relatively short preparation time for fabrication of small number of different parts, and the high "freedom of form" that is important in the early stages of the design process. This is the reason that many designers choose them as a tool for prototyping. Depending on the type of the prototype, the appropriate process from the range of the AM processes can be chosen [3]. This information has to be determined in the early stages of the design process. This is why the adequate knowledge of DfAM has become critical for successfully fabricated parts.

There are numerous publications in the area of DfAM, since this is one of the key challenges concerning AM [4]. Lack of knowledge in DfAM can prevent the designers of fully exploiting the possibilities of AM [5] and generally desirable the AM technologies to reach their full potential in the industry [6,7].

Teaching design for additive manufacturing is well suited to problem-based learning [8] and the learning through real-world problems is far greater than from just memorizing course material. The authors present the application of DfMA to improve the design from both functional and technological aspects. AM Design Principle Cards are proposed by Hwang [9] as a design tool to support innovative thinking around Additive Manufacturing in the early stages of the design process. The cards are divided into four categories to support efforts related to products, business processes, design processes, and printing principles. Prabhu R et al. [10] divide the DfAM principles into opportunistic and restrictive and conclude that teaching both DfAM concepts resulted in the generation of ideas with greater AM technical goodness. They suggest that future research must investigate how DfAM education affects the design outcomes at different stages of the design process. They notice that the differences in prior experience could influence the motivation toward learning about DfAM. In a different study [11] the results from a training workshop for DfAM show that the students do not apply DfAM knowledge in their design concepts. However, this study shows that appropriate knowledge about DfAM can help the students to be more willing to use these new technologies.

At "Ss Cyril and Methodius" University, Faculty of Mechanical Engineering in Skopje, courses computer-aided design, 3D modelling, and design prototyping are part of the undergraduate curriculum for industrial design studies. The courses in advanced product modelling, product customization and rapid prototyping, are offered at master's studies. In this paper, three study modules and their implementation in the learning process are discussed: (1) Parametric modelling methods, (2) Design for additive manufacturing (DfAM) and (3) Prototyping with AM technology.

Wider availably of AM technologies increased the popularity of parametric modelling in industrial design to create parts with complex features, shape, structure, and pattern and to gain unusual aesthetic attraction in a flexible way. Based on the computer-aided design knowledge gained at undergraduate level, the functionality of visual programming is introduced to the students and practiced with examples. Creation of a parametric model or a 3D pattern, requires background knowledge in geometric modelling and algorithmic programming. The primary advantage of a parametric model is that it can be relatively easy to vary, by changing its parameters and also, by changing the geometric elements used to build it. Therefore, industrial designers can shorten the time required to create a new model by reusing existing parametric models.

The DfAM study module is aimed to increase the theoretical and practical knowledge that supports development of prototypes and products that can be fabricated using different AM technologies. Rules for increasing the economics of the AM, such as selection of most suitable orientation for the object, minimization of process time and material quantity, surface quality are studied.

The module for learning AM technologies for prototyping is related to the available AM processes, materials, and their application. The students practice fabrication of designed models using the process of material extrusion or more precisely fused filament fabrication (FFF) and Stereolithography (SLA). Case studies of use of AM technologies for prototype fabrication, and also for low volume production of parts, assemblies and molds, are analyzed. The examples include decorative objects, personalized products, jewelry, bio inspired designs and multi-material parts.

One of the goals in the learning process is to recognize and exploit the eventual synergic effect of combining parametric modelling and AM technologies for creation of prototypes of objects with extraordinary forms, especially high value-added products, and also, to determine the extent of involving the end user (customer) in the design process. This is presented in Fig. 1, where the red frame highlights that DfAM process starts in the design modelling phase and continues with preparation of the model. Parametric modelling is an appropriate solution for parts fabricated with AM, since in the design phase all the important parameters for the fabrication process can be implemented in the design. This means that at the end of the design phase, we have a model ready for fabrication, which speeds up the process and saves valuable resources.

However, there are several limitation of AM processes, regarding appearance, such as the staircase effect on the surface, as the models are built by applying the material in layers. The working manner also affects the micro structure, resulting in anisotropy of material. Additionally, fabricating parts with overhangs and bridges requires use of support structures which adds time in the fabrication, post processing operations, and also waste material.

## 3. DfAM – Design rules and considerations

In order to get more relevant information, we conducted a survey among our alumni students, who attended the faculty before implementing the DfAM in our curriculum. We analyzed the problems when working with AM, stated by them (the students) and divided them into the following groups [13]:

- Un-fit technology,
- Inappropriate design,
- Inappropriate preparation for fabrication,
- Other factors.

Un-fit technology **-** The first thing to know is that these are not comprehensive technologies and that in some cases it is simply not smart and cost effective to use them. AM technologies should be avoided when production of large series is needed or series of products with simple design. These technologies are recommended for: parts with complex geometry, personalized parts, custom parts, and unique parts in small series. Furthermore, the availabil-

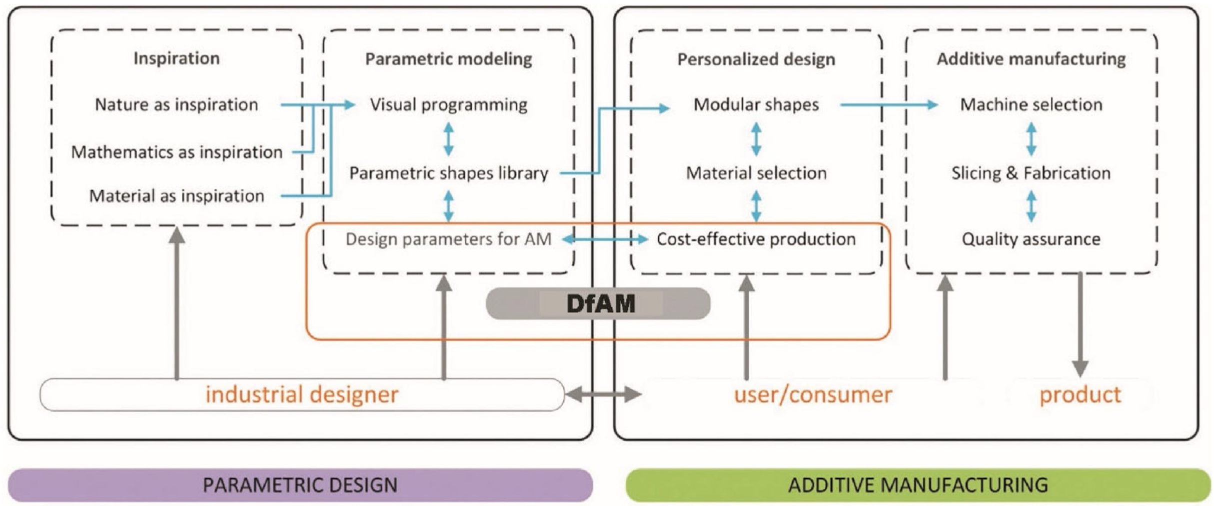



**Fig. 1.** Learning process for introduction of DfAM knowledge for economic fabrication of parametrically designed models with AM (Adapted from [12]).

ity of materials that can be applied depends on the choice of technology, and thus the surface quality and mechanical properties.

Inappropriate design – When designing a part to be made with AM technologies, i.e. specifically with the FFF process, it should be known that the complexity in the form does not affect either the time of production or the cost.

Inappropriate preparation for fabrication – the preparation for the preparation of the part greatly affects the final result that will be obtained. In open-source machines, the manipulation of the working parameters is completely left to the user, so if he does not have appropriate knowledge in the field, it will significantly affect the design.

Other factors - these include problems that may be caused by a mechanical problem with the machine (filament break, filament jam, etc.), or an electrical problem (power outage). When it comes to open-source machines, which do not have enclose the quality of the fabricated part is influenced by the external factors such as changes in room temperature.

These problems can be overcome by appropriate design for AM, appropriate application of process parameters and visual aspects in the DfAM.

### 3.1. Appropriate design for AM

As it is stated before in the text and in the ISO / ASTM standardization [3], in the early stages of the design process it should be decided if the AM is the most appropriate technology, regarding the complexity, number of parts, materials. The part dimensions are not recommended to excide the build envelope. The part should be designed to avoid the use of support structures. Large flat surfaces should be avoided in order to ensure better part quality. Filet and chamfers should be considered on the bottom surface to ensure easier part removal. Polygonal mesh should be used for the designing because the margin of error in the conversion to polygonal model (stl) is smaller [14]. When designing individual elements with a small cross section, the deviation in the profile should be expected. Additionally, for smaller deviations in the profile of the cylindrical elements, the orientation of the model during construction is of great importance. If the cross section is circular, it should be with an axis set at 90° in relation to the build plate.

In addition to the design rules and guidelines we have designed a benchmark model (Fig. 2) that can be used for evaluation of the FFF machine and examine its possibilities. The model integrates all the crucial elements in the FFF fabrication process, such us: thin features, inclined features, bridges, and text. The model should be fabricated without the use of support structures, and the positive outcome is presented in Fig. 2-b.

### 3.2. Appropriate application of process parameters

Positioning of the part on the build plate does not affect the quality of the part [15]. On the other hand, the orientation of the part to the build plate has a great impact on its quality. The influence of orientation is especially noticeable in parts with cylindrical elements. The cylindrical elements, positioned with the axis perpendicular to the build plate, have smaller dimensional and geometric deviations in relation to the parallel position of the axis. Accordingly, in the case of prismatic models, the surfaces placed perpendicular to the work surface during the fabrication has the best surface quality and the smallest deviations from the basic profile. The use of supporting material should be avoided whenever possible, as it causes a number of problems. The use of support structures, in addition to the poor quality of the surface after its removal, negatively affects other parameters, such as: extension of the fabrication time, increasing the amount of material used and additional time for post-processing (removal of supporting material and possible finishing of the model). In cases with complex, organic shapes there is no possibility to avoid the support structures and attempts should be made to minimize their impact. This is done by optimizing the geometry, the placement of the part and the way of generating the supporting material. In processes of extrusion of material that can work with multiple materials simultaneously, it is possible to use a water-soluble material for the support structures. In this way, the fabrication time is longer, but there is no negative impact on the surface quality. In those cases, there is no need to eliminate the use of supporting material, it is even recommended to use it because those machines are set up for such operation. The infill pattern and density should be wisely chosen since they affect the fabrication time and on the mechanical properties of the part. For models that need to perform responsible operations, it is recommended to use 100 % infill, also in cases where the model needs to be machined, e.g. to form openings. For most models used as appearance prototypes or souvenirs, it is sufficient to use a 20 % infill, sometimes even less.

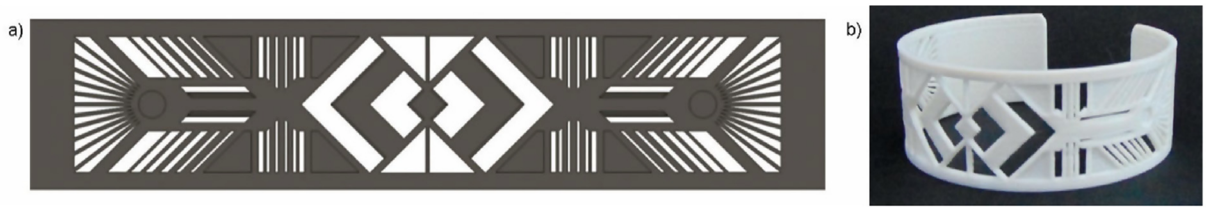



**Fig. 2.** Benchmark model – bracelet: (a) unfolded CAD model; (b) fabricated model [13].

### 3.3. Visual aspects in the DfAM

Main concern for the industrial designers is the appearance of the parts especially in the prototyping phase. One of the problem that can occur are not fabricated details. Therefore, in the slicing process, it should be checked whether the model can be fully fabricated. In case of models with thin walls and small details, the smallest possible layer thickness should be used. In cases of parts with patterns and small details, effort should be made to place them parallel to the build plate and use to smallest nozzle. If the patterns and details are placed at 90° in relation to the build plate, the layer height has the greatest impact on the output quality. The output quality of the parts fabricated with FFF is generally satisfactory. One concern is the warping of the part, this is why large thin parts should be avoided. As mentioned before one element concerning the appearance that cannot be omitted is the staircase effect. This effect can be diminished by using lower layer heights. It should be noted that on the inclining surfaces, especially with low angle according to the build plate, the stair case effect is visible, so here the orientation of the part should be revised. Inevitable high staircase effect is noted in parts with organic shape, so in this cases postproduction for smoothing the surface is advised. The appearance of threads, known as stringing, is a common occurrence in the manufacture of FFF parts. If the design is such that there are multiple elements with a small cross section at a small distance from each other, the effect of the stringing is even more visible. The working principle of FFF, as well as other AM processes, comes down to applying material in layers, one on top of the other. In cases when material has to be applied on a certain layer, and there is no material or support structures underneath, the sagging or drooping occurs. When parts with large overhangs or bridges without support structures needs to be fabricated, rotation of the part should be considered. By rotating the part 90°, the horizontal elements will become vertical and it will be built without problem. Having to use support structures one should know that their removal leaves visible marks on the part, so the part should be finished with some chemical or mechanical process.

## 4. Case studies

Parametric modelling in industrial design is used to distinguish, represent and modify the important geometric aspects of the product design, such as shape, pattern, structure, modules and their combination. Parametric models can be reused and recomposed for creating new product models. Parametric models can be created using the functionality of a CAD system, or a programming language that is connected to the CAD system. For industrial designers, one of the most interesting features of parametric models are flexible patterns. Patterns created by algorithms are related to industrial manmade products. Patterns are part of the aesthetic language, and add rhythm, order and texture to the objects. The pattern building elements can have geometric or organic shape.

Industrial designer can take the role of both user or creator of parametric models. Variation of existing parametric designs and reuse of program modules are the most common operations to develop a design model and also a model for AM in a limited time. Because of the simplicity, the variation of parametric models has been used also to personalize some aspects of product design by involving of the customer. The process for parametric modelling using Rhinoceros/Grasshopper, can begin with reuse of existing parametric models, continue with gradual increase of the programming skills by combining of existing or new programming modules, to arrive to creating models with new algorithms. The approach is based on case studies, to gain wider understanding in short time period and to raise interest among students. In the following, the main aspects of the learning process are illustrated with examples.

The presented examples are part of the teaching material for the students and give insight into the part of the teaching methodology and also the DfAM guidelines that are considered in during parametric modelling. Understanding and analogy to the model examples that are previously prototyped with the available AM technologies can support the design process. New models are created by variation of the design parameters, basic geometry and exchange or replacement of compatible building blocks, without changing the structure of the algorithm. Selected examples have potential for creating many variants, even unexpected. Each example is accompanied by a set of DfAM rules. The AM process is simulated with the AM software. At this stage, the students learn about the use of parameters to drive design changes, model structure and transformations. Creativity in such process and the value of the designs can be limited by the underling examples. More creative and demanding examples are developed through projects.

### 4.1. Example 1 – Decorative flower pot

Common materials used are ceramics, clay, molded plastics, and sheet metal. It can be manufactured as a highly creative unique item, but also with mass production processes as low-cost item. There is interest for product prototyping, customization and personalization. The model of a flower pot, shown in Fig. 3 is used to study the common elements of a parametric model, such as the underlying shape (surface of the flower pot), pattern elements (pyramids) and the pattern matrix (rectangular grid of points), used to morph the pattern on the surface. With the change of the model parameters of the shape and pattern, many variants of the pot can be created. This example is used to study algorithm structure, geometric and mathematical elements that constitute the model. Basic AM guidelines for standard FFF process, used in the modeling phase for this example, are: (1) Decide feasible part orientations for prototyping, (2) Compare the dimensions of the model with the dimensions of the machine workspace, (3) Keep angles to the normal of the machine plate to <45°, (4) Avoid supports, (5) Minimize material content.

### 4.2. Example 2 – Jewelry and box series

Products of this type are mostly from metals, plastics and also in many other materials. These are highly creative items, produced in

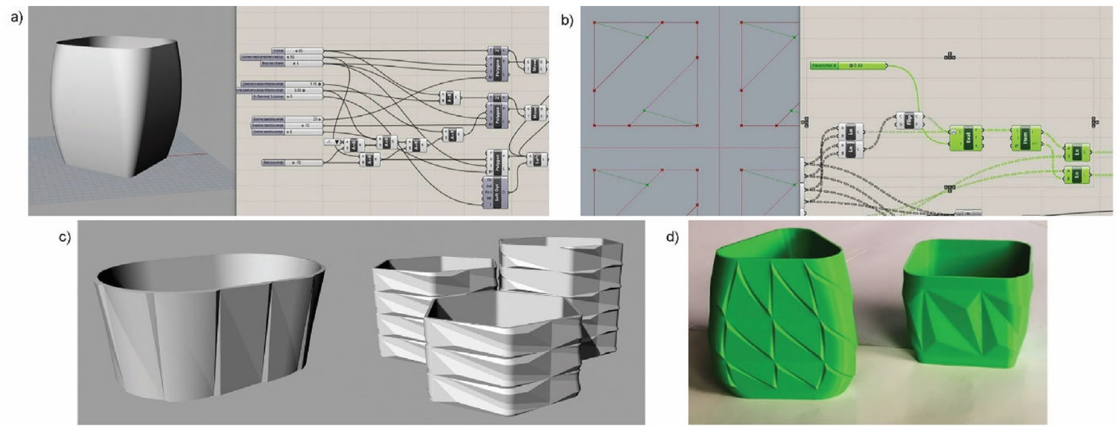



**Fig. 3.** Elements of a parametric model for flower pot decorated with 3D pattern: (a) parametric shape; (b) pattern element with pyramids; (c) variants of the model; and (d) pot 1 and pot 2 prototypes.

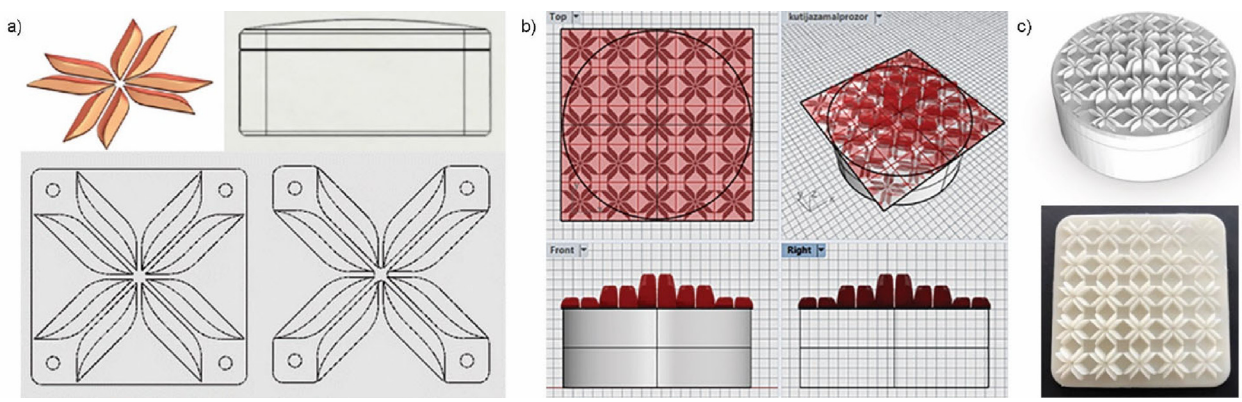



**Fig. 4.** Decoration of box lid with a 3D pattern of geometric elements: (a) pattern element and operations for creation of the pattern; (b) attractor point in the center for changing the height of the pattern; (c) lid prototype with attractor point in the corner of the pattern.

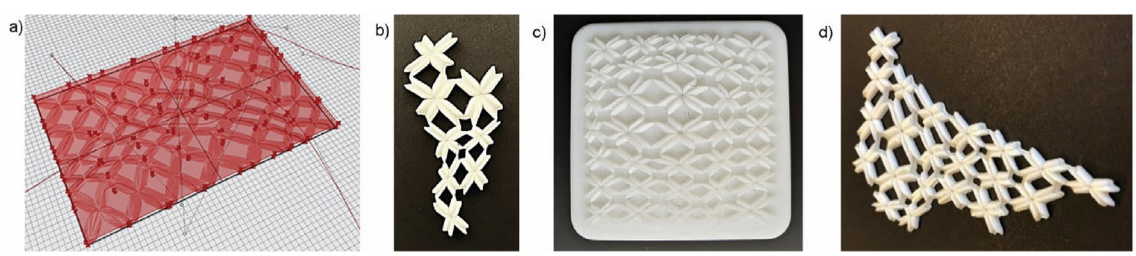
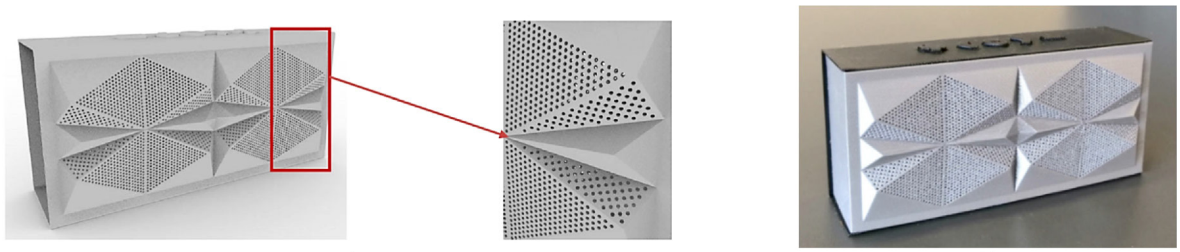



**Fig. 5.** Creating series: (a) deformation of the pattern using attractor curves; (b) earing prototype; (c) box lid; (d) neckless design.

limited quantities, with attention to small details. There is high interest for product prototyping, customization and personalization. Box lid with a 3D pattern is shown in Fig. 4 and earing and neckless in Fig. 5. The decorative geometric element is created in a CAD tool and imported in STL format. The pattern is created using a rectangular matrix and the morph function. The height of the pattern can be modified using an attractor point, such as in Fig. 4 (b, c). Basic AM guidelines for standard SLA process that are used in the modelling phase for this example, are: (1) Decide feasible part orientations for prototyping. (2) Use orientation with less layers, (3) Compare the dimensions of the model with the dimensions of the printing space, (4) Keep angles to the normal of the machine plate to < 45°, (5) Avoid supports, (6) Select layer thickness considering time vs quality of surface and details.

### *4.3. Example 3 – Bluetooth speaker case*

This type of objects is produced from sheet metal, molded plastic, and wood. It is a design and technical item, produced in limited up to large quantities. There is interest for product customization. In Fig. 6, the design of a Bluetooth speaker case is shown, where the pattern of small holes is placed on the faces of the origami style pattern composed of large pyramids. The speaker case is created using computer-aided design software. A prototype of the speaker case is required to examine shape aesthetic features, evaluate ergonomics and interaction. The basic model of the case is divided in two parts to avoid supports and to access the interior. Basic AM guidelines for standard FFF process that are used in the modelling phase are the same as in Example 1.



a) b)

**Fig. 6.** Bluetooth speaker case: (a) pattern of holes on the faces of a pattern of pyramids in origami style; (b) prototype using FFF.

**Table 1**
Printing parameters for the analyzed examples in Figs. 3–6.

| Example | Process | Material | Orientation | Layer [mm] | Infill | Weight | Build time |
|---|---|---|---|---|---|---|---|
| Pot 2 | FFF | PLA | Vertical | 0.2 | 20 % | 85 gr. | 10 h |
| Bluetooth speaker case – front | FFF | PLA | Vertical | 0.2 | No infill | 18 gr. | 7 h. |
| Bluetooth speaker case - back | FFF | PLA | Horizontal | 0.2 | No infill | 27 gr. | 11 h. |
| Earing | SLA | resin | Horizontal | 0.05 | NA | 8 ml | 30 min |
| Box lid | SLA | resin | Horizontal | 0.05 | NA | 13.7 ml | 38 min |

The Libraries of algorithms that can be used as building blocks are available in the literature and as add-in software modules that improve the parametric capabilities and algorithms for designing creative concepts. Development of new parametric models and algorithms can be improved further by training in computer programming, and also by involving computer science experts in the design process. More advanced examples are studied through different student projects to gain more knowledge and skills in parametric modeling, application of DfAM rules and prototyping with AM. The work on projects stimulates students' creativity and results in design ideas that get more advantage from using AM technologies.

For prototyping, the model can be in different scale, the parts can be combined together and other simplifications can be done to save time and to limit the evaluation to only few aspects of the designed object. The manufacturing process and material that are used for prototyping, are usually different than those used for production. The parametric model can be examined and modified further according to the requirements of the production process as soon as it is determined. More details about the fabrication of the prototypes for the presented examples are given in Table 1.

## 5. Conclusion

Teaching method based on selected case studies, practical examples and design projects, helps in learning how to solve up-to-date design problems that involve concurrent application of parametric modelling, DfAM and prototyping with AM. The interrelations between three bodies of knowledge are recognized early in the design stage and synergy effects can be exploited in creating new designs that can be fabricated with minor preparation.

The learning process is supported by involving the students in creating variants of previously prepared examples of parametrically designed objects, accompanied by the DfAM guidelines and the prototyping experience with AM. The learning continues with work on more creative design projects to develop and prototype new parametrically modelled objects, and with improved understanding of the capabilities of AM processes. The proposed learning methodology is aimed to encourage the industrial design students to prototype their designs early in the design process, to be able to evaluate and present their designs in short time. The positive experience with AM fabrication comes from understanding how the part has been manufactured and also, knowing the successful applications of these technologies in the design and industrial practice.

## CRediT authorship contribution statement

**Tatjana Kandikjan:** Supervision, Visualization, Writing – original draft, Data curation, Validation, Conceptualization, Methodology. **Jelena Djokikj:** Visualization, Writing – original draft, Resources, Investigation, Validation, Methodology, Conceptualization. **Ile Mircheski:** Visualization. **Elena Angeleska:** Project administration, Writing – review & editing.

## Data availability

No data was used for the research described in the article.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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